

Connections

In what follows, k is a commutative ring, and A is a commutative k -algebra.

We denote P to be $A \otimes_k A$, and P^1 the first order neighborhood of A in P , and I the kernel of $P \rightarrow A$ which is generated by $1 \otimes a - a \otimes 1$. Then $P^1 = P/I^2$. We sometimes use $\mathbf{1}$ to denote the element $1 \otimes 1$.

There are two maps

$$\begin{aligned} d_0: A &\rightarrow P^1, a \mapsto a \otimes 1 \\ d_1: A &\rightarrow P^1, a \mapsto 1 \otimes a \end{aligned}$$

We thus have a truncated cosimplicial diagram

$$\begin{array}{ccc} & d_0 & \\ A & \xrightarrow{\quad} & P^1 \\ & d_1 & \end{array}$$

We regard d_0 as a left A -module structure on P^1 and d_1 as a right A -module structure. If E is an A -module, we write $P^1 \otimes_A E$ (resp. $E \otimes_A P^1$) to denote the P^1 -module $d_1^* E$ (resp. $d_0^* E$), and unless otherwise specified, its A -module structure means $d_{0*} d_1^* E$ (resp. $d_{1*} d_0^* E$).

Let E be an A -module. A connection of E over k is a k -linear map

$$\nabla: E \rightarrow E \otimes_A \Omega_{A/k}^1$$

such that

$$\nabla(fs) = f\nabla s + s \otimes df$$

for $s \in E$, $f \in A$. We denote modules with connections by MC .

Proposition 1.1. *The following data are equivalent:*

- A connection ∇ on the module E ,
- An A -linear homomorphism $\theta: E \rightarrow E \otimes_A P^1$ that is id_E modulo Ω^1 ,
- A P^1 -linear homomorphism $\epsilon: P^1 \otimes_A E \rightarrow E \otimes_A P^1$ that is id_E modulo Ω^1 . Any such homomorphism is automatically an isomorphism

Sketch of proof. Given a connection ∇ , let $\theta = \nabla + \text{id}_E \otimes d_0$ which is a d_1 -linear map from E to $E \otimes_A P^1$. By extension of scalars, we get the morphism

$$\epsilon: P^1 \otimes_A E \rightarrow E \otimes_A P^1.$$

Since ϵ is the identity map modulo I , we know ϵ itself is an isomorphism. $\square$

Remark 1.2. The isomorphism ϵ may be interpreted as a “first order parallel transport”. Suppose k is a field, $X = \text{Spec}(A)$, and x be a rational point on X . Let v be a (Zariski) tangent vector of X at x . Such tangent vector should be thought of as an infinitesimal path. Algebraically, v means a morphism $A \rightarrow k[\epsilon]$. We denote this morphism by v and the constant map to x also by x .

$$\begin{array}{ccc} E' & \longrightarrow & E \\ \downarrow & & \downarrow \\ \text{Spec}(k[\epsilon]) & \xrightarrow[v]{v} & X \end{array}$$

Let E be a vector bundle with a connection on X . Since the map (v, x) to $X \times_k X$ factors through P^1 , we get an isomorphism $v^*E \simeq x^*E$. However, x^*E is the trivial bundle, so ϵ gives a way of trivializing the bundle E along the (infinitesimal) path v .

Such picture coincides with differential geometry. In differential geometry, given a connection ∇ on vector bundle E over X , we can parallel transport elements on E along paths in X . Since parallel transport induces isomorphism between fibers, it gives a way of trivializing the vector bundle E along paths. The connection can be recovered from parallel translation as follows. For a path $\gamma: [a, b] \rightarrow M$, we denote $P_{t_0, t}^\gamma$ the parallel transport from $E_{\gamma(t_0)}$ to $E_{\gamma(t)}$. Suppose $\gamma(t_0) = x$ and $\gamma'(t_0) = v$, then for any section s of E ,

$$\nabla_v s(x) = \left. \frac{d}{dt} \right|_{t=t_0} (P_{t_0, t}^\gamma)^{-1} s(\gamma(t)).$$

So the connection can be recovered from the first order parallel transport. This coincides with our algebraic calculation. Indeed, for a section $s \in E$, suppose $\nabla s = \sum s_i \otimes dx_i$. We have

$$\begin{aligned} \epsilon: P^1 \otimes E &\longrightarrow E \otimes P^1, \\ \mathbf{1} \otimes s &\longmapsto s \otimes \mathbf{1} + \sum s_i \otimes (x_i \otimes 1 - 1 \otimes x_i). \end{aligned}$$

Therefore after pulling back to $k[\epsilon]$, we get

$$\begin{aligned} (v, x)^* \epsilon: v^*E &\longrightarrow x^*E, \\ \mathbf{1} \otimes s &\longmapsto s \otimes \mathbf{1} + \sum s_i \otimes (v(x_i) - x(x_i)) = s \otimes \mathbf{1} + \nabla_v s \end{aligned}$$

The final equality follows from that $v - x$ is precisely the derivation D_v corresponding to the tangent vector v .

The category of connections

An A -linear map $f: (E, \nabla_E) \rightarrow (F, \nabla_F)$ is called horizontal if $f\nabla_E = \nabla_F f$. It is equivalent to either of the following commutative diagrams:

$$\begin{array}{ccc} E & \xrightarrow{\theta_E} & E \otimes P^1 \\ f \downarrow & & \downarrow f \otimes 1 \\ F & \xrightarrow{\theta_F} & F \otimes P^1 \end{array} \quad \begin{array}{ccc} P^1 \otimes E & \xrightarrow{\epsilon_E} & E \otimes P^1 \\ 1 \otimes f \downarrow & & \downarrow f \otimes 1 \\ P^1 \otimes F & \xrightarrow{\epsilon_F} & F \otimes P^1 \end{array}$$

MC admit tensor products. Let (E, ∇_E) and (F, ∇_F) be modules with connections, then we can define a connection on $E \otimes F$ by

$$\nabla(s \otimes t) = \nabla_E s \otimes t + s \otimes \nabla_F t. \quad (1)$$

Proposition 2.1. *In terms of ϵ , the tensor product connection is given by*

$$P^1 \otimes E \otimes F \xrightarrow{\epsilon_E \otimes 1} E \otimes P^1 \otimes F \xrightarrow{1 \otimes \epsilon_F} E \otimes F \otimes P^1$$

Proof. Let $s \in E, t \in F$ and we shall consider the image of $1 \otimes s \otimes t$. Suppose $\nabla s = \sum s_i \otimes \omega_i$, then

$$(\epsilon_E \otimes 1)(1 \otimes s \otimes t) = s \otimes 1 \otimes t + \sum s_i \otimes \omega_i \otimes t.$$

By the definition of ϵ_F and that Ω^1 is square-zero,

$$(1 \otimes \epsilon_F)(s_i \otimes \omega_i \otimes t) = s_i \otimes \omega_i(t \otimes 1 + \nabla t) = s_i \otimes t \otimes \omega_i.$$

Therefore, we have

$$(1 \otimes \epsilon_F) \circ (\epsilon_E \otimes 1)(1 \otimes s \otimes t) = s \otimes t \otimes 1 + s \otimes \nabla t + s_i \otimes t \otimes \omega_i. \quad (2)$$

Comparing (1) and (2) the result follows. $\square$

Let B be a commutative A -algebra. A structure of precrysals on B over A is an isomorphism as P^1 -algebras

$$P^1 \otimes B \simeq B \otimes P^1$$

which reduces to identity modulo Ω^1 .

Proposition 2.2. *A structure of precrysals on B is equivalent to a left splitting of the exact sequence*

$$B \otimes_A \Omega_{A/k} \longrightarrow \Omega_{B/k} \longrightarrow \Omega_{B/A} \longrightarrow 0.$$

Proof. In particular, B admits a connection. That ϵ preserves structure of algebra is equivalent to the commutativity of the following diagram

$$\begin{array}{ccc} P^1 \otimes B \otimes B & \xrightarrow{\epsilon_{B \otimes B}} & B \otimes B \otimes P^1 \\ 1 \otimes m_B \downarrow & & \downarrow m_B \otimes 1 \\ P^1 \otimes B & \xrightarrow{\epsilon_B} & B \otimes P^1 \end{array}$$

In terms of connections, this is equivalent to saying

$$\nabla(bb') = b\nabla b' + b'\nabla b.$$

Therefore, the map $\nabla: B \rightarrow B \otimes \Omega_{A/k}^1$ is a k -derivation. Thus, we get a left splitting of the exact sequence

$$B \otimes_A \Omega_{A/k} \longrightarrow \Omega_{B/k} \longrightarrow \Omega_{B/A} \longrightarrow 0.$$

Conversely, given a left splitting we yield a k -derivation on B as well as an A -connection, which gives a precystal structure on B . $\square$

Everything we discussed above may be globalized to schemes. Let S be a k -scheme. We denote $D(1)$ the first order infinitesimal neighborhood of the diagonal $S \rightarrow S \times_k S$. We have a truncated simplicial diagram of schemes

$$D(1) \begin{array}{c} \xrightarrow{p_0} \\ \leftarrow \Delta \rightarrow \\ \xrightarrow{p_1} \end{array} S.$$

If E is a quasicoherent sheaf on S , a connection on E relative to k is an isomorphism of $D(1)$ -modules $\epsilon: p_0^*E \simeq p_1^*E$ that is id_E after reduction to diagonal. If X is a scheme over S , a structure of precystal on X is an isomorphism $D(1)$ -schemes $\epsilon: p_0^*X \simeq p_1^*X$ that is id_X after reduction to diagonal. Equivalently, it is a left splitting of the short exact sequence

$$f^*\Omega_{S/k} \longrightarrow \Omega_{X/k} \longrightarrow \Omega_{X/S} \longrightarrow 0.$$

We define $T_{S/k} := \text{Spec}(\text{Sym } \Omega_{S/k})$. From the universal property of symmetric algebra and differential module, we know

$$T_{S/k}(Z) = S(Z[\epsilon]).$$

A precystal structure on X gives a section of the morphism of tangent scheme $T_{X/k} \rightarrow f^*T_{S/k}$. In terms of functor of point of view, it is described as follow: given a Z -point of f^*T_X which may be described the following commutative digrams with solid arrows:

$$\begin{array}{ccc} Z & \xrightarrow{t} & X \\ \downarrow & \nearrow & \downarrow \\ Z[\epsilon] & \xrightarrow{\delta} & S \end{array}$$

we want to find a point in $T_{Y/k}(Z)$, that is, a dashed arrow filling the diagram. Let

$$\iota: Z[\epsilon] \rightarrow Z \rightarrow Z[\epsilon] \xrightarrow{\delta} S$$

be the “constant” map induced by δ , then we have an isomorphism $(\iota, \delta)^* \epsilon: \iota^* X \simeq \delta^* X$. Now t gives a section $\alpha: Z[\epsilon] \rightarrow \iota^* X$, then composing the isomorphism $\iota^* X \simeq \delta^* X$ we get the desired map $Z[\epsilon] \rightarrow \delta^* X$.

$$\begin{array}{ccccc} \delta^* X & \xleftarrow{\simeq} & X[\epsilon] & \xrightarrow{\quad} & X \\ & \searrow \alpha & \downarrow \scriptstyle{\wr} & & \downarrow f \\ & & Z[\epsilon] & \xrightarrow{(\iota, \delta)} & D(1) \xrightarrow[p_1]{p_0} S \end{array}$$

Compared to the previous remark of first order parallel transport, we will call this first order horizontal lift.

Let us justify the first order horizontal lift is indeed given by the the splitting of exact sequence of differnetial forms. We may reduce everything to affine case, to assume $k = \text{Spec } k$, $S = \text{Spec } A$, $X = \text{Spec } B$ and $Z = \text{Spec } C$. Firstly, we shall note that the isomorphism

$$\text{Hom}_k(\text{Sym}_A \Omega_{A/k}, C) \simeq \text{Hom}_k(A, C[\epsilon])$$

should be interpreted as follows. Both sides are naturally isomorphic to the data of pairs (g, D) where g is a k -algebra morphism $g: A \rightarrow C$ and D is a derivation of A valued in C , whose A -module structure is given by g . Indeed, homomorphisms from A to $C[\epsilon]$ are of forms $g + \epsilon D$, and a homomorphism from $\text{Sym}_A(\Omega_{A/k})$ to C are determined by (g, D) where g is the map on degree 0 parts, and D is the map on degree 1 parts, mapping dx to Dx .

With this in mind, given a precystal structure $\nabla: B \rightarrow B \otimes_A \Omega_{A/k}$, the morphism $\Omega_{B/k} \rightarrow B \otimes \Omega_{A/k}$ is given by $db \mapsto \nabla b$. Given a commutative diagram with solid arrows

$$\begin{array}{ccc} B & \xrightarrow{t} & C \\ f \uparrow & \searrow \text{dashed} & \uparrow \\ A & \xrightarrow{(g, D)} & C[\epsilon] \end{array}$$

which is a C -point in $B \otimes_A \text{Sym}_A(\Omega_A)$, we seek for a dashed diagram filler which means a C -point of $\text{Sym}_B(\Omega_B)$. Suppose $\nabla b = \sum b_i \otimes da_i$, then under the morphism $\Omega_B \rightarrow B \otimes \Omega_A \rightarrow C$

$$db \mapsto \sum b_i \otimes da_i \mapsto \sum b_i Da_i. \quad (3)$$

On the other hand, denote $f_0 = (g, 0)$ and $f_1 = (g, D)$, then we need to understand the composed morphism

$$\begin{array}{ccccc} B & \longrightarrow & f_1^* B & \xrightarrow{\simeq} & f_0^* B \\ & & & \searrow & \downarrow t[\epsilon] \\ & & & & C[\epsilon] \end{array}$$

Under $P^1 \otimes B \simeq B \otimes P^1$, we have

$$\mathbf{1} \otimes b \longmapsto b \otimes \mathbf{1} + \sum b_i \otimes (1 \otimes a_i - a_i \otimes 1).$$

After pulling this back along (f_1, f_0) , we yield

$$1 \otimes b \longmapsto b \otimes 1 + \sum b_i \otimes (f_1(a_i) - f_0(a_i)) = b + \sum b_i D a_i \epsilon. \quad (4)$$

Comparing (3) and (4), we see they indeed coincide.